# <u>Climate Action Plan Proposal</u>

*(Aligned with the **Global Covenant of Mayors for Climate and Energy)***

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## 1. INTRODUCTION

### 1.1. Location and Overview

**Baramati** is a small city in Maharashtra located in the Pune district. With a population of around ***70,000*** (*as per recent estimates*), it serves as a key agricultural and industrial hub. Its economy is driven by sugarcane farming, textile manufacturing, and small-scale industries.

### 1.2. City Profile

- **Location**: Pune District, Maharashtra, India
- **Population**: ~70,000
- **Economy**: Primarily agricultural, with industrial contributions from sugar and textile production.

### 1.3. Purpose of the Climate Action Plan

To reduce greenhouse gas (GHG) emissions, enhance climate resilience, and improve the city's sustainability by addressing its environmental challenges through targeted and achievable actions.

### 1.4. Environmental Challenges

- ***Air Quality***: Vehicular emissions and industrial pollution contribute to deteriorating air quality. The sudden rise in population has led to the chase of vehicular services at an alarming rate which has led to a deteriorating trend of the air-quality of Baramati.
- ***Waste Management***: Inefficient segregation and improper disposal of municipal waste leads to environmental hazards. Also, it creates an overall unhygienic overview of the city whilst also taking up valuable space.
- ***Water Scarcity***: Overuse of groundwater for agriculture and irregular rainfall patterns create water shortages. The heavy usage of chemicals has also led to leaching of toxins to the groundwater causing it to degrade as well. The disruptions of the standard rainfall patterns have adversely impacted the entire agricultural sector that requires immediate action as it may otherwise lead to a food security issue if not resolved.
- ***Dependence on Non-Renewable Energy Sources***: Baramati's municipal operations and growing energy demands rely heavily on non-renewable energy sources, contributing to significant greenhouse gas (GHG) emissions. The lack of widespread adoption of renewable energy, such as solar and wind power, results in higher energy costs and environmental degradation.

- **Declining Urban Green Cover and Loss of Biodiversity**: Rapid urbanization has led to the depletion of green spaces and habitats in Baramati, negatively impacting air quality, biodiversity, and the city's resilience to climate change.
- **Vulnerability of Agriculture to Climate Change**: Agriculture, the backbone of Baramati's economy, faces significant threats from erratic rainfall, droughts, and depleting soil health due to unsustainable practices. These challenges not only jeopardize crop yields but also contribute to rising emissions through inefficient resource use.

## 2. BASELINE ASSESSMENT (*as per GCOM standards*)

### 2.1. Emissions Inventory

*i.* Transportation: 45%
*ii.* Industrial activities: 30%
*iii.* Waste: 15%
*iv.* Other sources: 10%

### 2.2. Vulnerability Assessment

*i.* Water scarcity due to declining groundwater levels and irregular rainfall.
*ii.* Increased health risks from air and waste pollution.

## 3. VISION AND GOALS

### 3.1. Vision Statement

"Baramati will become a model Tier-3 city in Maharashtra by 2030, demonstrating sustainable urban growth through efficient resource management, renewable energy adoption, and climate-resilient infrastructure."

### 3.2. Targets

1. **Emission Reduction**: Achieve a ***30% reduction*** in GHG emissions by ***2030***
2. **Water Conservation**: Reduce agricultural water use by ***40%*** through sustainable practices by ***2028***.
3. **Energy transition to renewables**: Achieve **50%** renewable energy use in municipal operations by **2028**.
4. **Climate-resilient agriculture**: Enhance agricultural productivity by **20%** while reducing emissions by **30%** by **2030**.
5. **Urban greening efforts**: Increase urban green cover by **30%** by **2030** to enhance air quality and biodiversity.
6. **Waste Management**: Segregate and recycle ***70%*** of municipal waste by ***2026***.
7. **Sustainable Mobility**: Transition ***20%*** of city commuters to public and non-motorized transport by ***2030***.

## 4. WHY A CLIMATE ACTION PLAN IS NEEDED

A **Climate Action Plan (CAP)** is a strategic framework designed to address and mitigate the impacts of climate change at a local, regional, or national level. It outlines specific actions and policies to reduce greenhouse gas (GHG) emissions, adapt to climate-related challenges, and promote sustainability. can help Baramati address pressing environmental issues and prepare for future challenges, including:

- ***Reducing Pollution***: Improving air and water quality will enhance public health and ecosystem stability.
- ***Resource Management***: Efficient waste and water management can preserve resources for future generations.
- ***Economic Growth***: Sustainable practices can attract investments and create ***green jobs***, boosting the local economy.
- ***Capacity Building***: Adoption of planned actions in view of climate change will lead for expansion of the ***Adaptive Capacity*** of the stakeholders of the city of Baramati.

Without intervention, these challenges could escalate, affecting the city's liveability and economic stability.

The ***International collective action*** against the cause of Climate Change mitigation acts as the driving factor among others for the need for a Climate Action Plan. Preparing for the fast-approaching adversities due to the man induced ***tipping-points*** infringements. Adoption of the ***Sustainable Development Goals*** in view of the development of the plan holds grave importance.

## 5. FOCUS AREAS FOR THE PLAN

### 5.1. Focus Area 1: Renewable Energy Transition

#### 5.1.1. *Key Goal*:

Achieve **50%** renewable energy use in municipal operations by **2028**.

#### 5.1.2. *Strategic Actions*:

*i.* **Solar Rooftop Systems**:

- ✓ Install solar panels on government buildings, schools, and hospitals.
- ✓ Provide subsidies to encourage residential solar adoption.

*ii.* **Microgrids for Rural Areas**:

- ✓ Establish solar-powered microgrids for nearby villages to reduce dependency on traditional electricity grids.

*iii.* **Public-Private Partnerships (PPPs)**:

- ✓ Partner with renewable energy firms to implement large-scale solar and wind projects.

**5.1.3.** ***Expected Outcomes***:

- Significant reduction in energy costs for municipal services.
- Lesser fossil-fuel burning to generate energy.
- **25%** of Baramati's electricity demand met by renewable sources by **2030**.

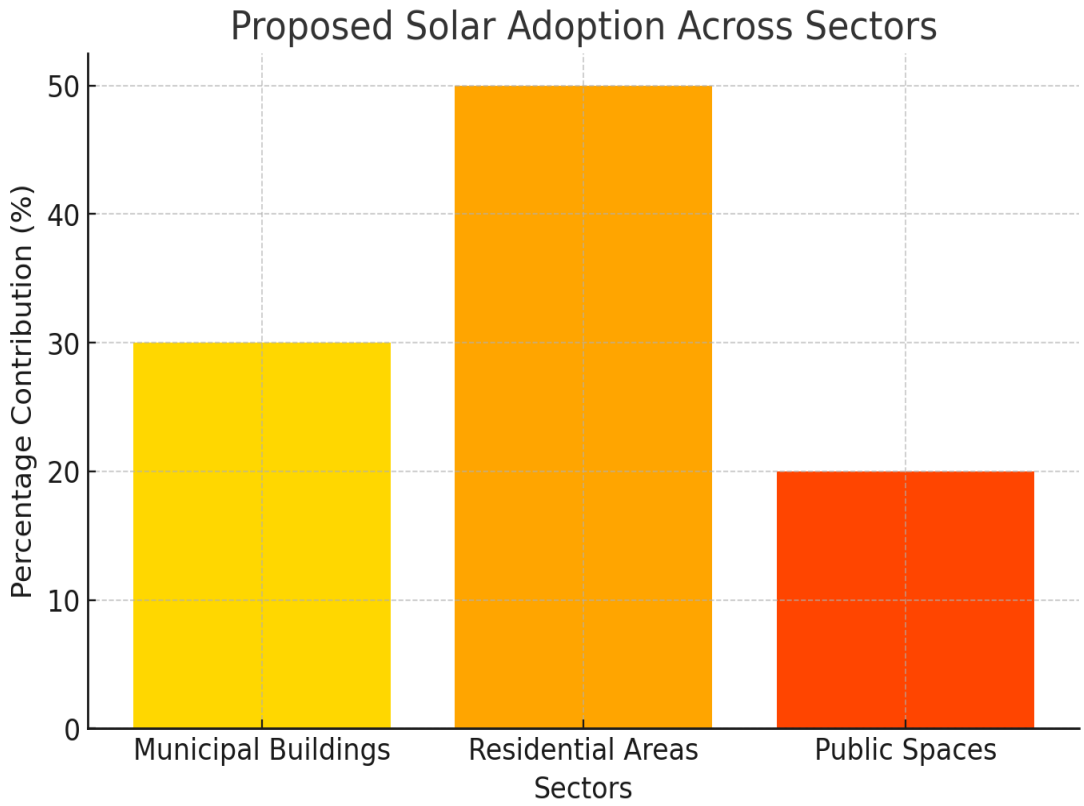



***Chart 1. Line chart representing the proposed Solar Adoption***

**5.2. Focus Area 2: Urban Greening and Biodiversity**

**5.2.1.** ***Key Goal***:

Increase urban green cover by **30%** by **2030** to enhance air quality and biodiversity.

**5.2.2.** ***Strategic Actions***:

*i.* **Urban Afforestation**:

- ✓ Plant **10,000** native tree species along roads, parks, and public spaces over **five years**.
- ✓ Collaborate with schools for tree-planting drives.

*ii.* **Biodiversity Corridors**:

- ✓ Develop green corridors linking urban parks and nearby forested areas.
- ✓ Protect local flora and fauna through habitat restoration projects.

*iii.* **Green Infrastructure Development**:

- ✓ Encourage green rooftops and vertical gardens in urban buildings.
- ✓ Provide tax incentives for developers implementing green infrastructure.

### 5.2.3. *Expected Outcomes*:

- Improved air quality by reducing particulate matter and CO2 levels.
- Increased resilience to urban heat and flood events.

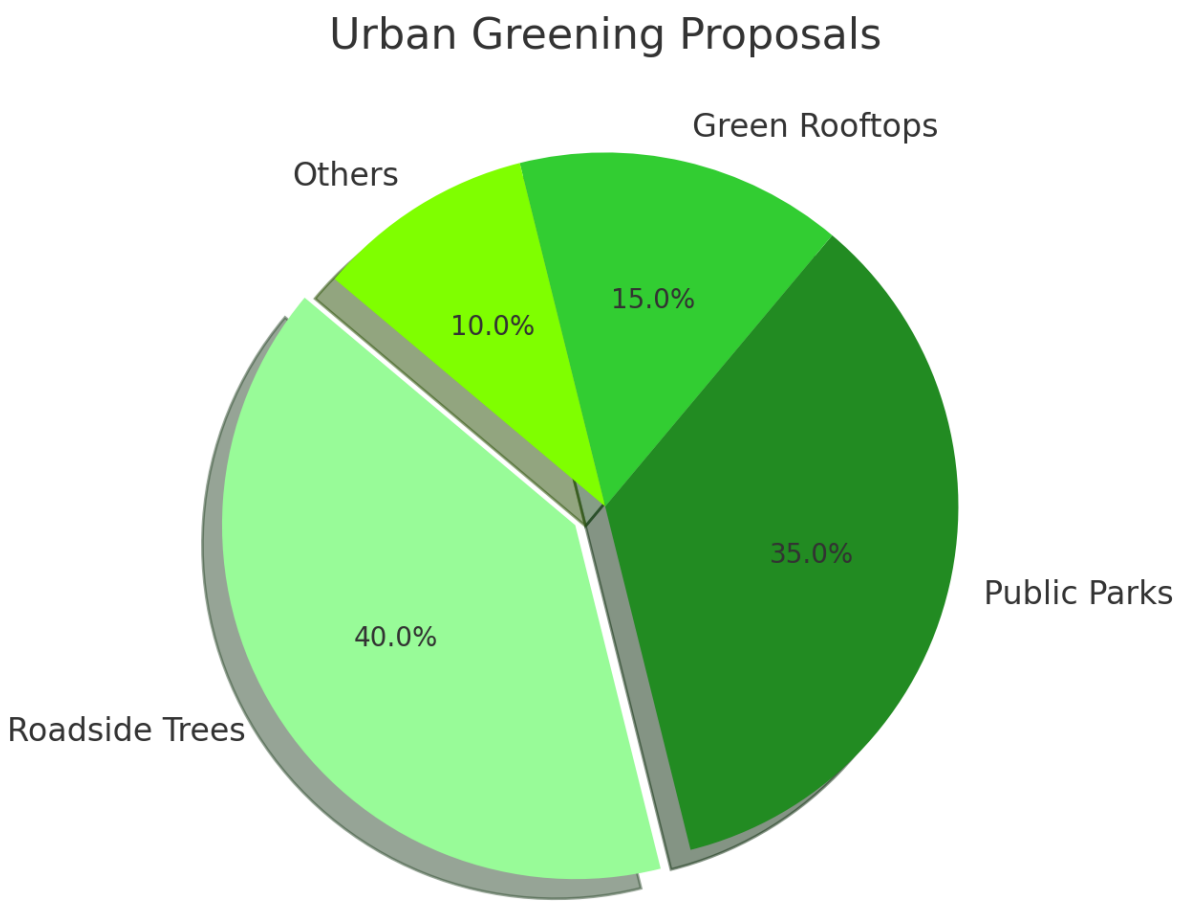



***Chart 2. Pie-chart depicting the proposed Urban greening***

### 5.3. Focus Area 3: Climate-Resilient Agriculture

#### 5.3.1. *Key Goal*:

Enhance agricultural productivity by **20%** while reducing emissions by **30%** by **2030**.

#### 5.3.2. *Strategic Actions*:

*i.* **Climate-Smart Farming Practices**:

- ✓ Promote crop rotation, intercropping, and soil conservation techniques.
- ✓ Train farmers in agroforestry and sustainable pest management.

*ii.* **Agri-Tech Adoption**:

- ✓ Deploy weather stations and IoT-based monitoring tools for precision farming.
- ✓ Provide mobile apps for farmers to access weather forecasts and market prices.

*iii.* **Low-Emission Livelihoods**:

- ✓ Support small-scale biogas plants for rural households.
- ✓ Encourage the use of solar pumps for irrigation.

#### 5.3.3. **Expected Outcomes**:

- Reduced vulnerability to drought and erratic rainfall.
- Increased farmer incomes through higher productivity and resource efficiency.

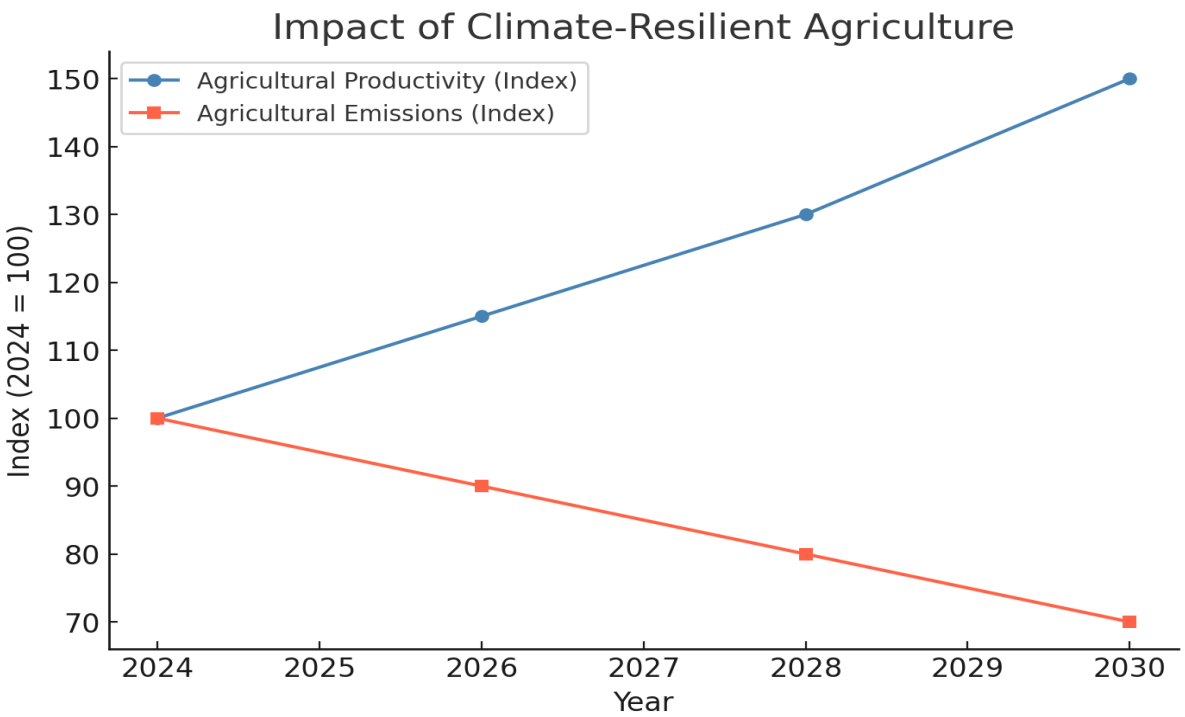



***Chart 3. Line-chart depicting the Impacts of Climate-resilient Agricultural practices***

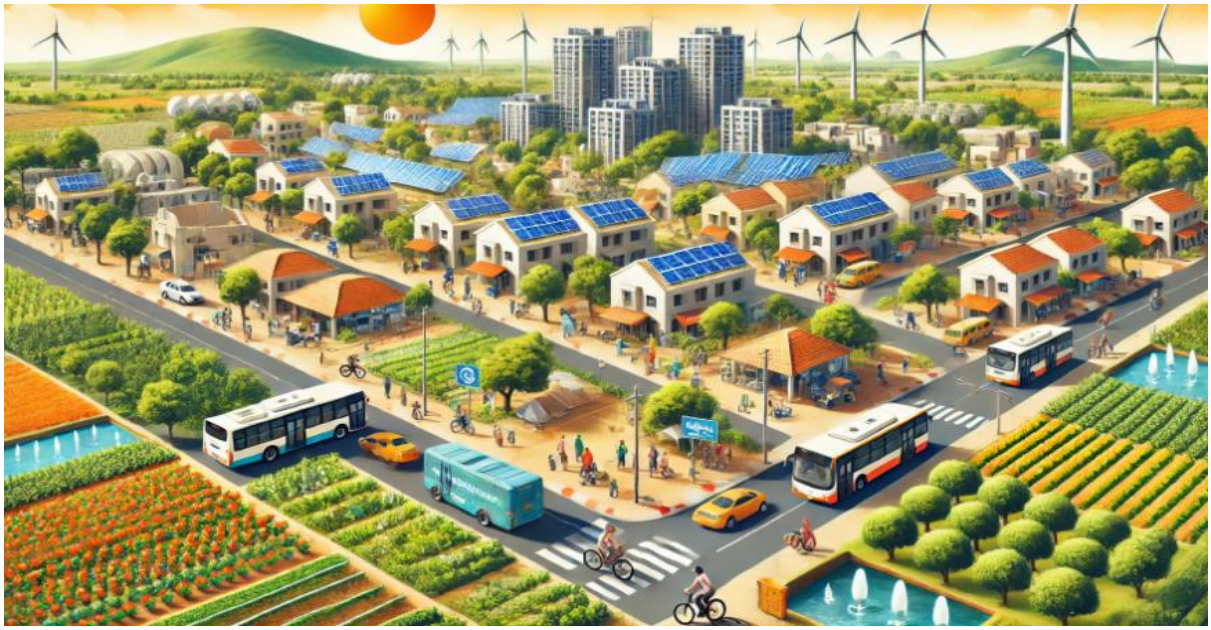
## 6. CONCLUSION

The ***Climate Action Plan for Baramati*** outlines a comprehensive strategy to address the city's environmental challenges and foster sustainability. By improving ***waste management*** through segregation, recycling, and awareness campaigns, the plan aims to reduce landfill waste, curb methane emissions, and promote a circular economy. ***Water conservation measures***, such as drip irrigation, rainwater harvesting, and community water literacy initiatives, tackle the critical issue of groundwater depletion, ensuring sustainable agricultural practices and alleviating water scarcity.

***Sustainable mobility solutions***, including the introduction of e-buses, cycling lanes, and car-free days, aim to reduce vehicular emissions and improve air quality while encouraging low-carbon commuting habits. The ***transition to renewable energy***, through solar adoption and microgrids, decreases reliance on fossil fuels, cutting greenhouse gas emissions and promoting energy equity. ***Urban greening efforts***, such as planting native trees, creating biodiversity corridors, and integrating green infrastructure, enhance air quality, mitigate urban heat, and support local biodiversity.

The plan also emphasizes ***climate-resilient agriculture*** by promoting smart farming techniques, renewable energy use in agriculture, and the adoption of precision technologies. These measures aim to improve productivity, reduce emissions, and build resilience to climate variability. To ensure progress, the plan incorporates data-driven monitoring tools like GIS mapping and IoT devices for tracking key environmental metrics.

Collectively, these actions position Baramati to achieve significant environmental improvements, including a ***30% reduction in greenhouse gas emissions by 2030***, while enhancing resilience to climate impacts. The plan fosters equity by engaging diverse stakeholders and serves as a model for sustainable urban growth in other Tier-3 cities across India. Through these targeted and collaborative efforts, ***Baramati can transform into a resilient, sustainable city while contributing to global climate goals***.



***Picture. AI generated image based on the recommendations suggested above (GPT-4o)***

## 7. REFERENCES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